

WATER AND CARBON CYCLE IN NATURE - FULL MIND MAP

Topic-wise | Hinglish | SSC / BPSC / BSSC / Railway

Topics in this chapter:

1. Carbon Cycle
2. Water Cycle & Solar Energy
3. Transplantation & Root Hairs

1. Carbon Cycle

- Carbon cycle mein respiration, organic matter ka decay aur volcanic action CO₂ add karte hain. Respiration organic compounds todkar CO₂ release karta hai; decomposition carbon atmosphere mein return karta hai; volcanoes Earth interior se CO₂ release karte hain. [Q1]
- Photosynthesis CO₂ add nahi karta; plants atmospheric CO₂ ko carbohydrates banane mein use karte hain aur oxygen release karte hain. Isliye photosynthesis carbon dioxide ko atmosphere se remove karta hai. [Q2]

2. Water Cycle & Solar Energy

- Water cycle ka main energy-driver Sun hai, gravity ke saath. Solar energy oceans/water bodies se evaporation aur land/vegetation se evapotranspiration karwati hai; cooling par condensation, phir gravity se precipitation hota hai. [Q3]
- Rainfall ke liye atmospheric humidity moisture source provide karti hai. Moist air rise karke cool/saturate hoti hai, condensation hota hai aur droplets grow karke rain ban sakte hain. High humidity akeli heavy rain guarantee nahi karti - uplift, cooling, winds aur cloud microphysics bhi matter karte hain. [Q4]

3. Transplantation & Root Hairs

- Transplanted seedlings aksar isliye fail hote hain kyunki uprooting mein fine roots/root hairs damage ya lose ho jaate hain. Root hairs water aur dissolved minerals absorb karne ke liye surface area badhate hain; loss se absorption kam, transpiration continue aur transplant shock/wilting ho sakta hai. [Q5]